

Python Basic Data Types -

1. Integer
2. float
3. Complex
4. Boolean
5. String

1] Integer (int) -

- An Integer is whole number. -
positive, negative or zero - written without any decimal point.
- Python's int type has no fixed size limit. So it can represent extremely large whole no. as long as your computer's memory allows it.

Integer can be expressed in four way.

1. Binary - base 2

2. Decimal - base 10 $\rightarrow$ Human can understand decimal

3. Octal - base - 8

4. Hexadecimal - base - 16

Decimal — 10 (Human) ✓

binary — 1010

octal — 12

Hexadecimal — A

decimal_num = 10

binary_num = 0b1010

octal_num = 0o12

hexadecimal_num = 0xA

4
way
to
represent

23 float -

A float represents a real number that has a fractional (decimal) part.

Python can express a float either in the usual decimal notation or in notation .
exponential

1] decimal notation -

$$p_i = 3.14$$

2] exponential notation -

$$\text{var1} = 6.02e23$$

$\Downarrow$

$$6.02 \times 10^{23}$$

3] Complex -

Complex number has two parts

1) real part

2) imaginary part

ex. $Var1 = 5 + 3j$

↑ ↖
real part imaginary part

4] Boolean (bool) -

- The boolean type has only two possible values. True & False.
- Booleans are mainly produced as the result of a comparison.

marks = 78

78 > 35

has_passed = marks > 35

print(has_passed) → True

marks = 27

27 > 35

has_passed = marks > 35

print(has_passed) → False

5] String - A string is an ordered, immutable sequence of unicode characters.

- String can be written using single quote ('), double quote(") or tripple quote(''', ''')

eg.

City = 'Pune'

Course = "Artificial Intelligence"

Course_desc = " " "This is very
descriptive AI Course " " "

